AI-Powered Career Guidance & Job Apply Agent

Bridging the Employment Gap with Agentic AI (SDG 4 & 8)

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The Challenge: Employment Gap in India

- **Skill Mismatch:** Graduates often struggle to identify job roles that accurately match their specific technical skill sets.
- **Manual Inefficiency:** Manual job searching and generic applications lead to low response rates and candidate fatigue.
- **Counseling Gap:** Students in Tier-2 and Tier-3 cities often lack access to professional career placement support.
- **SDG Alignment:**
  - **SDG 4 (Quality Education):** Promoting skill-based career development.
  - **SDG 8 (Decent Work):** Optimizing the transition from education to employment.

The Solution: Intelligent Career Orchestration

- **End-to-End Automation:** A "One-click" system that manages the full application lifecycle.
- **Functional Workflow:**
  - **Resume Analysis:** Automated extraction of skills and experience via AI.
  - **Dynamic Job Matching:** Real-time alignment with market opportunities.
  - **Professional Personalization:** Automated drafting of formal business cover letters.
  - **Direct Dispatch:** Smart Gmail integration for immediate HR outreach.

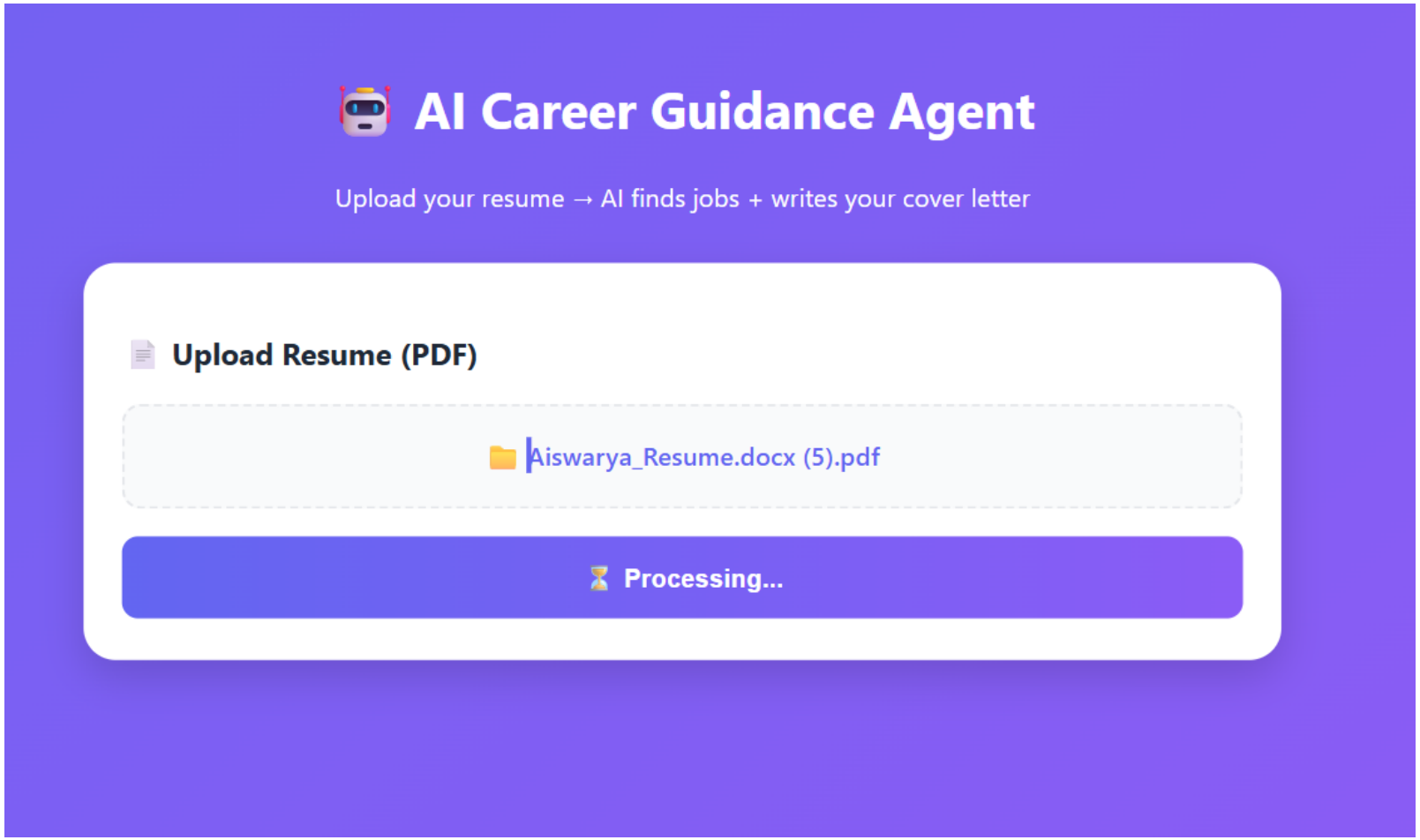
Technical Architecture & Stack

Project Tech Stack

- **Frontend:** React.js (Vite) for a responsive user interface.
- **Backend:** Python + FastAPI for high-performance logic.
- **AI Workflow:** LangGraph for multi-step agent orchestration.
- **Model:** Llama 3.3 (via Groq API) for fast, accurate reasoning.

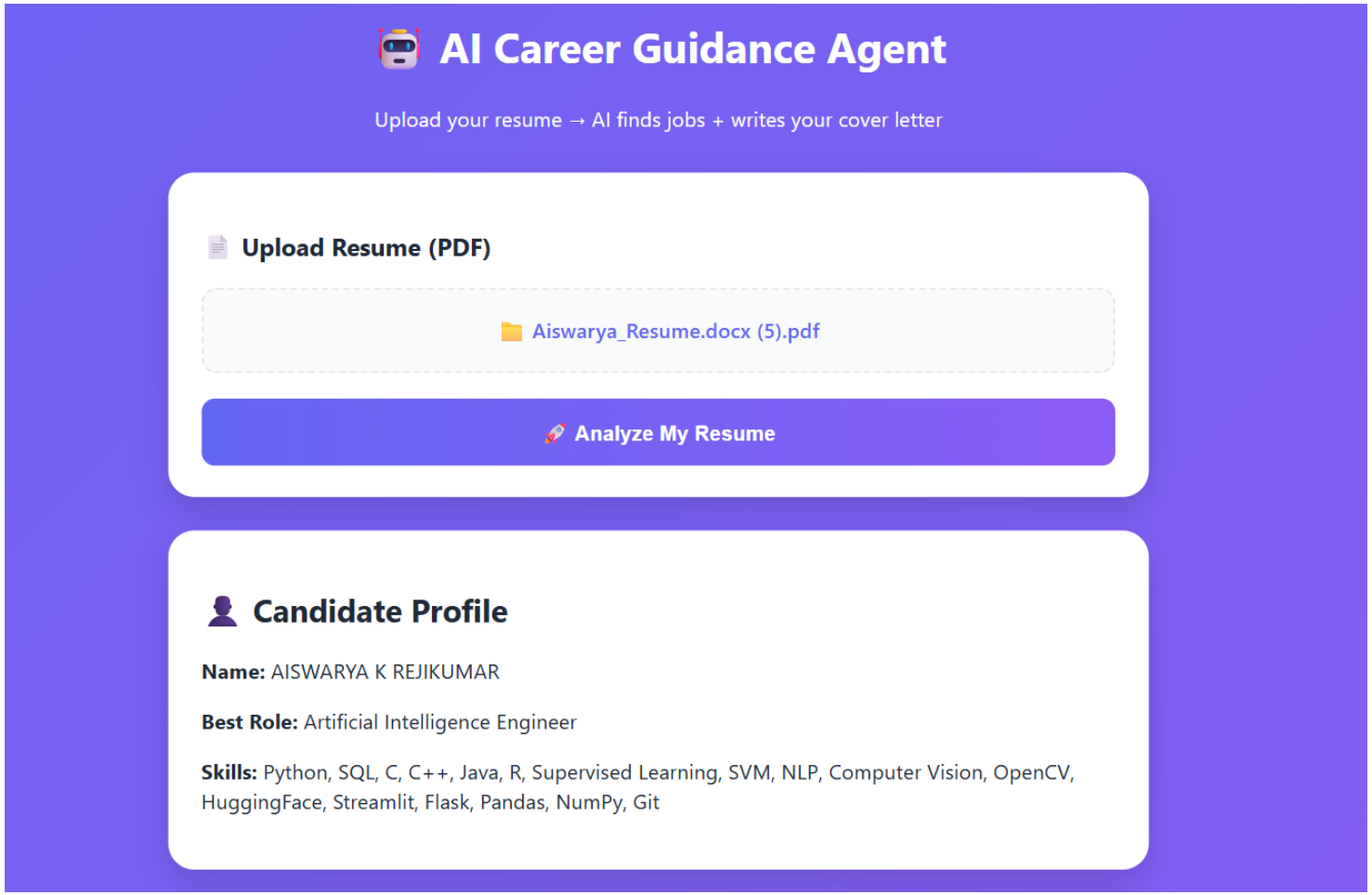
- **State Management:** LangGraph ensures reliable transitions between data extraction and job search nodes.

Implementation: Minimalist User Interface



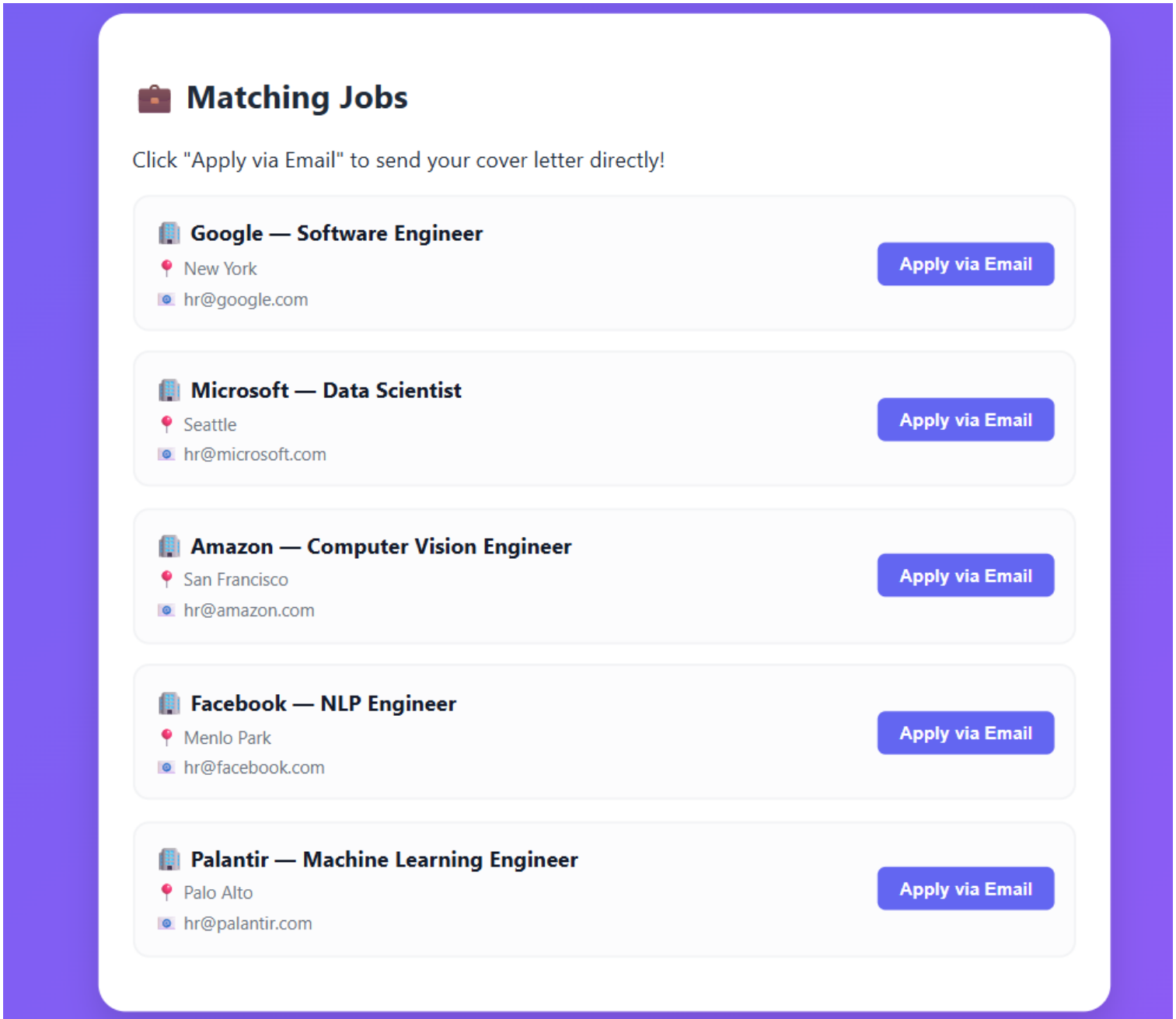
- **Design:** Clean, intuitive UI built with React and Tailwind for high accessibility.
- **Interaction:** Simple PDF upload functionality allowing for instant backend processing.

Implementation: Automated Resume Analysis

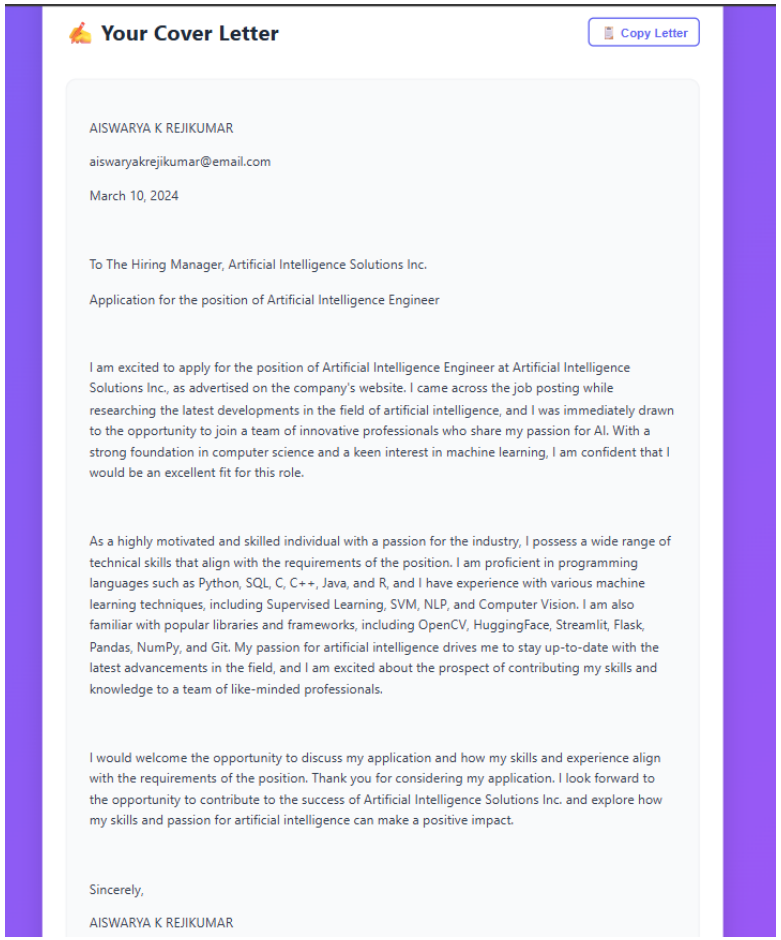


- **Parsing:** Agentic extraction of Candidate Name, Best Role, and Technical Skills.
- **Output:** Provides a structured profile used to drive the matching engine.

Implementation: Real-Time Job Matching



Implementation: Automated Cover Letter Generation



✍️ **Your Cover Letter**

Copy Letter

AISWARYA K REJIKUMAR

aiswaryakrejikumar@email.com

March 10, 2024

To The Hiring Manager, Artificial Intelligence Solutions Inc.

Application for the position of Artificial Intelligence Engineer

I am excited to apply for the position of Artificial Intelligence Engineer at Artificial Intelligence Solutions Inc., as advertised on the company's website. I came across the job posting while researching the latest developments in the field of artificial intelligence, and I was immediately drawn to the opportunity to join a team of innovative professionals who share my passion for AI. With a strong foundation in computer science and a keen interest in machine learning, I am confident that I would be an excellent fit for this role.

As a highly motivated and skilled individual with a passion for the industry, I possess a wide range of technical skills that align with the requirements of the position. I am proficient in programming languages such as Python, SQL, C, C++, Java, and R, and I have experience with various machine learning techniques, including Supervised Learning, SVM, NLP, and Computer Vision. I am also familiar with popular libraries and frameworks, including OpenCV, HuggingFace, Streamlit, Flask, Pandas, NumPy, and Git. My passion for artificial intelligence drives me to stay up-to-date with the latest advancements in the field, and I am excited about the prospect of contributing my skills and knowledge to a team of like-minded professionals.

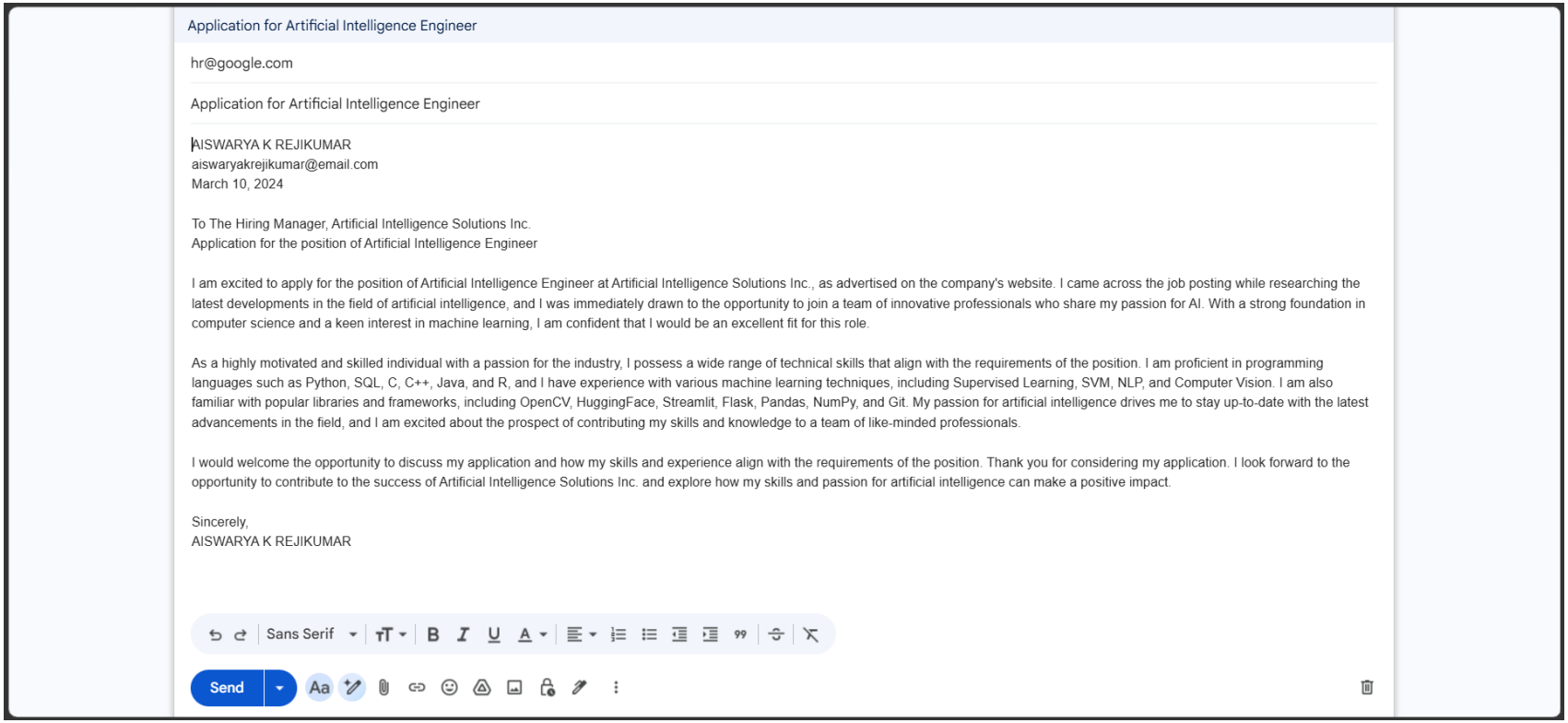
I would welcome the opportunity to discuss my application and how my skills and experience align with the requirements of the position. Thank you for considering my application. I look forward to the opportunity to contribute to the success of Artificial Intelligence Solutions Inc. and explore how my skills and passion for artificial intelligence can make a positive impact.

Sincerely,

AISWARYA K REJIKUMAR

- **Personalization:** Generates formal business letters tailored to the specific job role and user skills.
- **Standardization:** Follows professional formatting to increase candidate response rates.

Final Step: Seamless Gmail Integration



- **Integration:** One-click Gmail redirection for final application dispatch.
- **Automation:** Pre-fills HR email, Subject line, and Body for immediate professional use.

Conclusion & Future Roadmap

- **Conclusion:** Successfully developed a functional Agentic AI framework to bridge the career guidance gap for Indian students.
- **Future Scope:**
  - Implementation of persistent user history tracking using Supabase.
  - Expansion of fully automated background workflows via N8N.
  - Supporting regional Indian languages to improve inclusivity and reach.